

## **Internship Task 1:**

### **Network Traffic Analysis Report using Wireshark**

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**Task Title:** Capture and Analyze Network Traffic Using Wireshark

## 1. Introduction

Network traffic analysis is the process of capturing, inspecting, and interpreting network packets to understand communication occurring between devices. This exercise was performed using Wireshark, a widely used network protocol analyzer, to capture live traffic and identify commonly used network protocols.

The objective of this task was to capture network packets, identify multiple protocols, analyze their behavior, and understand how data flows through a network.

## 2. Objective

The primary objectives of this task were:

- Capture live network traffic using Wireshark.
- Identify at least three different network protocols.
- Analyze packet details and protocol behavior.
- Understand how network communication occurs.
- Document findings and observations.

## 3. Tools Used

| Tool         | Purpose                     |
|--------------|-----------------------------|
| Wireshark    | Packet capture and analysis |
| Kali Linux   | Operating System            |
| Web Browser  | Generate web traffic        |
| Ping Utility | Generate ICMP traffic       |
| nslookup     | Generate DNS traffic        |

#### **4. Methodology**

The following steps were performed:

1. Wireshark was launched on Kali Linux.
2. The active network interface (wlan0) was selected.
3. Live packet capture was started.
4. Various network activities were generated:
  - a. Browsing websites such as Google, GitHub, and YouTube.
  - b. Executing DNS queries using nslookup.
  - c. Sending ICMP packets using ping.
5. Packet capture was stopped after several minutes.
6. Protocol filters were applied to isolate specific traffic types.
7. Packet statistics were analyzed.
8. Findings were documented.

## 5. Protocols Identified

The following protocols were identified during packet analysis:

## 5.1 DNS (Domain Name System)

- *Purpose:*

DNS *translates human-readable domain names into IP addresses.*

- *Example Observed:*

A DNS query was generated when accessing google.com.

- *Observation:*

The client system sent a DNS request to the configured DNS server, which responded with the IP address of the requested domain.

- *Packet Count:*

## 214 Packets

**taskcapng**

| No. | Time         | Source        | Destination   | Protocol | Length | Info                                                                                                                |
|-----|--------------|---------------|---------------|----------|--------|---------------------------------------------------------------------------------------------------------------------|
| 135 | 13.871017727 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0x41a1 A google.com                                                                                  |
| 136 | 13.871133396 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0xb6bb AAAA google.com                                                                               |
| 137 | 13.117240834 | 10.164.126.36 | 10.164.126.97 | DNS      | 166    | Standard query response 0x41a1 A google.com A 142.250.143.101 A 142.250.143.100 A 142.250.143.102 A 142.250.143.103 |
| 138 | 13.135917636 | 10.164.126.36 | 10.164.126.97 | DNS      | 182    | Standard query response 0xb6bb AAAA google.com AAAA 2404:6800:4000:100c::64 AAAA 2404:6800:4000:100d::64            |
| 141 | 13.211891119 | 10.164.126.97 | 10.164.126.36 | DNS      | 132    | Standard query 0xbcae PTR 4.6.0.0.0.0.0.0.0.0.0.0.0.c.0.0.1.0.0.0.4.0.0.8.6.4.0.4.2.ip.arpa                         |
| 142 | 13.307856364 | 10.164.126.36 | 10.164.126.97 | DNS      | 166    | Standard query response 0xbcae PTR 4.6.0.0.0.0.0.0.0.0.0.0.0.c.0.0.1.0.0.0.4.0.0.8.6.4.0.4.2.ip.arpa                |
| 152 | 19.348404008 | 10.164.126.97 | 10.164.126.36 | DNS      | 71     | Standard query 0xd939 A chatgpt.com                                                                                 |
| 160 | 19.443941166 | 10.164.126.36 | 10.164.126.97 | DNS      | 103    | Standard query response 0xd939 A chatgpt.com A 172.64.155.209 A 104.18.32.47                                        |
| 168 | 19.443365591 | 10.164.126.97 | 10.164.126.36 | DNS      | 71     | Standard query 0x4126 AAAA chatgpt.com                                                                              |
| 169 | 19.611789974 | 10.164.126.36 | 10.164.126.97 | DNS      | 99     | Standard query response 0x4126 AAAA chatgpt.com AAAA 2606:4700:83b2:7cbc:c23b:4b5:5ff2:75cd                         |
| 176 | 26.457468293 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0xa2cb A google.com                                                                                  |
| 177 | 26.509539007 | 10.164.126.36 | 10.164.126.97 | DNS      | 166    | Standard query response 0xa2cb A google.com A 142.250.143.139 A 142.250.143.113 A 142.250.143.112 A 142.250.143.111 |
| 178 | 26.513285229 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0xb9ba AAAA google.com                                                                               |
| 179 | 26.519931822 | 10.164.126.36 | 10.164.126.97 | DNS      | 182    | Standard query response 0xb9ba AAAA google.com AAAA 2404:6800:4000:100c::66 AAAA 2404:6800:4000:100d::66            |
| 190 | 34.532526085 | 10.164.126.97 | 10.164.126.36 | DNS      | 69     | Standard query 0x7016 A amazon.com                                                                                  |
| 202 | 34.808016792 | 10.164.126.36 | 10.164.126.97 | DNS      | 288    | Standard query 0x7016 A amazon.com A 3.33.192.77 A 15.197.245.13 NS ns1.amzdnsns.org                                |
| 263 | 34.884051589 | 10.164.126.97 | 10.164.126.36 | DNS      | 69     | Standard query 0x3088 AAAA amazon.com                                                                               |
| 267 | 34.921872054 | 10.164.126.36 | 10.164.126.97 | DNS      | 137    | Standard query response 0x3088 AAAA amazon.com SOA dns-external-master.amazon.com                                   |
| 225 | 47.546633330 | 10.164.126.97 | 10.164.126.36 | DNS      | 88     | Standard query 0x6102 A profile.accounts.firefox.com                                                                |
| 226 | 47.546713411 | 10.164.126.97 | 10.164.126.36 | DNS      | 88     | Standard query 0x580a AAAA profile.accounts.firefox.com                                                             |
| 227 | 47.639297103 | 10.164.126.36 | 10.164.126.97 | DNS      | 212    | Standard query response 0x6102 A profile.accounts.firefox.com CNAME mozilla.map.fastly.net A 205.171.253.20         |
| 228 | 47.642693210 | 10.164.126.36 | 10.164.126.97 | DNS      | 224    | Standard query response 0x580a AAAA profile.accounts.firefox.com CNAME mozilla.map.fastly.net A 205.171.253.20      |
| 250 | 56.136711774 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0xc65a HTTPS github.com                                                                              |
| 251 | 56.132671733 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0x93a3 A github.com                                                                                  |
| 252 | 56.132942875 | 10.164.126.97 | 10.164.126.36 | DNS      | 70     | Standard query 0xf75a AAAA github.com                                                                               |
| 253 | 56.310920113 | 10.164.126.36 | 10.164.126.97 | DNS      | 86     | Standard query response 0x93a3 A github.com A 20.207.73.82                                                          |
| 254 | 56.310922123 | 10.164.126.36 | 10.164.126.97 | DNS      | 135    | Standard query response 0xf75a AAAA github.com SOA dns1.p08.nsonone.net                                             |
| 255 | 56.310921441 | 10.164.126.36 | 10.164.126.97 | DNS      | 135    | Standard query response 0xc65a HTTPS github.com SOA dns1.p08.nsonone.net                                            |

```

Frame 135: Packet 70, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_95:c5:9a (00:e1:8c:95:c5:9a), Dst: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15)
Internet Protocol Version 4, Src: 10.164.126.97, Dst: 10.164.126.36
User Datagram Protocol, Src Port: 46899, Dst Port: 53
Domain Name System (query)
```

Packets: 20685 - Displayed: 214 (1.0%) - Dropped: 0 (0.0%) Profile: Default

## 5.2 TCP (Transmission Control Protocol)

- *Purpose:*

TCP provides reliable, connection-oriented communication between devices.

- *Example Observed:*

TCP packets were observed during HTTPS communication with websites.

- *Observation:*

TCP packets included SYN, SYN-ACK, and ACK packets that established connections before data transmission.

- *Packet Count:*

18353 Packets

task1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp

| No. | Time         | Source                 | Destination            | Protocol | Length | Info                                                                                |
|-----|--------------|------------------------|------------------------|----------|--------|-------------------------------------------------------------------------------------|
| 71  | 5.304351875  | 2606:4700:83b2:7cbc... | 2401:4900:7c8e:5de6... | TLSv1.2  | 117    | Application Data                                                                    |
| 72  | 5.304409406  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=347 Win=62 Len=0 TSval=1856619164 TSecr=4152925926  |
| 73  | 5.304446951  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=394 Win=62 Len=0 TSval=1856619164 TSecr=4152925926  |
| 74  | 5.304464090  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=425 Win=62 Len=0 TSval=1856619164 TSecr=4152925926  |
| 77  | 5.320623166  | 2606:4700:83b1:d8b7... | 2401:4900:7c8e:5de6... | TCP      | 86     | 443 → 55996 [ACK] Seq=104 Ack=117 Win=18 Len=0 TSval=3448744429 TSecr=3482501445    |
| 107 | 6.282294168  | 2606:4700:83b2:7cbc... | 2401:4900:7c8e:5de6... | TLSv1.2  | 1477   | Application Data                                                                    |
| 108 | 6.282295920  | 2606:4700:83b2:7cbc... | 2401:4900:7c8e:5de6... | TLSv1.2  | 315    | Application Data                                                                    |
| 109 | 6.282296189  | 2606:4700:83b2:7cbc... | 2401:4900:7c8e:5de6... | TLSv1.2  | 152    | Application Data, Application Data                                                  |
| 110 | 6.282391409  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=1816 Win=61 Len=0 TSval=1856620142 TSecr=4152926837 |
| 111 | 6.282438301  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=2045 Win=61 Len=0 TSval=1856620142 TSecr=4152926837 |
| 112 | 6.282450572  | 2401:4900:7c8e:5de6... | 2606:4700:83b2:7cbc... | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=2111 Win=61 Len=0 TSval=1856620142 TSecr=4152926837 |
| 131 | 7.613440616  | 2401:4900:7c8e:5de6... | 2606:4700:83b1:d8b7... | TLSv1.2  | 174    | Application Data                                                                    |
| 132 | 7.814886557  | 2606:4700:83b1:d8b7... | 2401:4900:7c8e:5de6... | TCP      | 86     | 443 → 55996 [ACK] Seq=104 Ack=205 Win=18 Len=0 TSval=3448746836 TSecr=3482503816    |
| 133 | 8.019319848  | 2606:4700:83b1:d8b7... | 2401:4900:7c8e:5de6... | TLSv1.2  | 165    | Application Data                                                                    |
| 134 | 8.062707384  | 2401:4900:7c8e:5de6... | 2606:4700:83b1:d8b7... | TCP      | 86     | 55996 → 443 [ACK] Seq=205 Ack=183 Win=63 Len=0 TSval=3482504266 TSecr=3448747094    |
| 172 | 25.430972030 | 2606:4700:83b1:d8b7... | 2401:4900:7c8e:5de6... | TLSv1.2  | 110    | Application Data                                                                    |
| 173 | 25.431054095 | 2401:4900:7c8e:5de6... | 2606:4700:83b1:d8b7... | TCP      | 86     | 55996 → 443 [ACK] Seq=205 Ack=207 Win=63 Len=0 TSval=3482521634 TSecr=3448764330    |
| 174 | 25.431420782 | 2401:4900:7c8e:5de6... | 2606:4700:83b1:d8b7... | TLSv1.2  | 114    | Application Data                                                                    |
| 175 | 25.632309181 | 2606:4700:83b1:d8b7... | 2401:4900:7c8e:5de6... | TCP      | 86     | 443 → 55996 [ACK] Seq=207 Ack=233 Win=18 Len=0 TSval=3448764627 TSecr=3482521634    |
| 182 | 30.956677622 | 2a03:2880:f379:120:... | 2401:4900:7c8e:5de6... | TLSv1.2  | 310    | Application Data                                                                    |
| 183 | 30.956787121 | 2401:4900:7c8e:5de6... | 2a03:2880:f379:120:... | TCP      | 86     | 35154 → 443 [ACK] Seq=70 Ack=296 Win=340 Len=0 TSval=3975833889 TSecr=3811622280    |
| 184 | 30.966514444 | 2401:4900:7c8e:5de6... | 2a03:2880:f379:120:... | TLSv1.2  | 155    | Application Data                                                                    |
| 185 | 31.006329314 | 2401:4900:7c8e:5de6... | 2a03:2880:f379:120:... | TLSv1.2  | 159    | Application Data                                                                    |
| 186 | 31.041270509 | 2a03:2880:f379:120:... | 2401:4900:7c8e:5de6... | TCP      | 86     | 443 → 35154 [ACK] Seq=296 Ack=139 Win=2038 Len=0 TSval=3811622532 TSecr=3975833898  |
| 187 | 31.082279713 | 2a03:2880:f379:120:... | 2401:4900:7c8e:5de6... | TCP      | 86     | 443 → 35154 [ACK] Seq=296 Ack=212 Win=2038 Len=0 TSval=3811622577 TSecr=3975833938  |
| 188 | 31.370313762 | 2a03:2880:f379:120:... | 2401:4900:7c8e:5de6... | TLSv1.2  | 157    | Application Data                                                                    |
| 189 | 31.410783265 | 2401:4900:7c8e:5de6... | 2a03:2880:f379:120:... | TCP      | 86     | 35154 → 443 [ACK] Seq=212 Ack=367 Win=340 Len=0 TSval=3975834343 TSecr=3811622747   |
| 210 | 36.490631464 | 2a03:2880:f379:120:... | 2401:4900:7c8e:5de6... | TLSv1.2  | 220    | Application Data                                                                    |

Frame 134: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlan0, id 0

Ethernet II, Src: Intel\_95:c5:9a (00:e1:8c:95:c5:9a), Dst: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15)

Internet Protocol Version 6, Src: 2401:4900:7c8e:5de6:d171:6e9:2b7:425, Dst: 2606:4700:83b1:d8b7:9a3b:662:ccf5:ac50

Transmission Control Protocol, Src Port: 55996, Dst Port: 443, Seq: 205, Ack: 183, Len: 0

0000 7a e3 b3 5c 3e 15  
0010 44 72 00 20 06 46  
0020 06 e9 02 b7 04 25  
0030 00 02 c0 f5 ac 50  
0040 04 eb 80 10 00 3f  
0050 cc 4a cd 8f b4 56

Transmission Control Protocol: Protocol

Packets: 20685 - Displayed: 18353 (88.7%) - Dropped: 0 (0.0%) Profile: Default

## 5.3 ICMPv6 (Internet Control Message Protocol)

- *Purpose:*

ICMPv6 is used for network diagnostics and error reporting.

- *Example Observed:*

ICMPv6 packets were generated using the ping command.

- *Observation:*

Echo Request and Echo Reply packets confirmed connectivity between hosts.

- *Packet Count:*

18 Packets

The image shows a Wireshark packet capture window titled 'task1.pcapng'. The filter is set to 'icmpv6'. The packet list shows 18 packets, all of which are ICMPv6 Echo (ping) requests and replies. The first packet (No. 139) is an Echo (ping) request from 2401:4900:7c8e:5de6 to 2404:6800:4000:100c. The subsequent packets are Echo (ping) replies from 2404:6800:4000:100c to 2401:4900:7c8e:5de6. The packet details pane shows the structure of the ICMPv6 Echo (ping) request, including the ID, sequence number, and hop limit. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 6 header, and ICMPv6 Echo (ping) request.

| No.   | Time         | Source              | Destination         | Protocol | Length | Info                                                                                           |
|-------|--------------|---------------------|---------------------|----------|--------|------------------------------------------------------------------------------------------------|
| 139   | 13.137518386 | 2401:4900:7c8e:5de6 | 2404:6800:4000:100c | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=1, hop limit=64 (reply in 140) (SendTTL=20, Round=21)       |
| 140   | 13.210196395 | 2404:6800:4000:100c | 2401:4900:7c8e:5de6 | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=1, hop limit=111 (request in 139) (SendTTL=20, Round=21)      |
| 143   | 14.138868462 | 2401:4900:7c8e:5de6 | 2404:6800:4000:100c | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 144) (SendTTL=20, Round=21)       |
| 144   | 14.223116600 | 2404:6800:4000:100c | 2401:4900:7c8e:5de6 | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=2, hop limit=111 (request in 143) (SendTTL=20, Round=21)      |
| 145   | 15.140397418 | 2401:4900:7c8e:5de6 | 2404:6800:4000:100c | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=3, hop limit=64 (reply in 146) (SendTTL=20, Round=21)       |
| 146   | 15.237755306 | 2404:6800:4000:100c | 2401:4900:7c8e:5de6 | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=3, hop limit=111 (request in 145) (SendTTL=20, Round=21)      |
| 147   | 16.142156940 | 2401:4900:7c8e:5de6 | 2404:6800:4000:100c | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=4, hop limit=64 (reply in 148) (SendTTL=20, Round=21)       |
| 148   | 16.241715046 | 2404:6800:4000:100c | 2401:4900:7c8e:5de6 | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=4, hop limit=111 (request in 147) (SendTTL=20, Round=21)      |
| 149   | 17.144192895 | 2401:4900:7c8e:5de6 | 2404:6800:4000:100c | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=5, hop limit=64 (no response found!) (SendTTL=20, Round=21) |
| 1648  | 54.382699970 | fe80::78e3:b3ff:fe5 | 2401:4900:7c8e:5de6 | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |
| 1649  | 54.382734791 | 2401:4900:7c8e:5de6 | fe80::78e3:b3ff:fe5 | ICMPv6   | 78     | Neighbor Advertisement 2401:4900:7c8e:5de6:d171:6e9:2b7:425 (sol)                              |
| 8629  | 78.476197418 | fe80::78e3:b3ff:fe5 | 2401:4900:7c8e:5de6 | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |
| 8630  | 78.476266126 | 2401:4900:7c8e:5de6 | fe80::78e3:b3ff:fe5 | ICMPv6   | 78     | Neighbor Advertisement 2401:4900:7c8e:5de6:d171:6e9:2b7:425 (sol)                              |
| 11535 | 85.341460730 | fe80::78e3:b3ff:fe5 | ff02::1             | ICMPv6   | 142    | Router Advertisement from 7a:e3:b3:5c:3e:15                                                    |
| 11536 | 85.351568445 | fe80::45a:b5ae:2fb5 | ff02::1             | ICMPv6   | 110    | Multicast Listener Report Message v2                                                           |
| 11572 | 85.754783297 | fe80::45a:b5ae:2fb5 | ff02::1             | ICMPv6   | 110    | Multicast Listener Report Message v2                                                           |
| 18852 | 98.931843473 | fe80::78e3:b3ff:fe5 | 2401:4900:7c8e:5de6 | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |
| 18853 | 98.931913512 | 2401:4900:7c8e:5de6 | fe80::78e3:b3ff:fe5 | ICMPv6   | 78     | Neighbor Advertisement 2401:4900:7c8e:5de6:d171:6e9:2b7:425 (sol)                              |

Frame 139: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface wlan0, id 0

Ethernet II, Src: Intel\_95:c5:9a (00:e1:8c:95:c5:9a), Dst: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15)

Internet Protocol Version 6, Src: 2401:4900:7c8e:5de6:d171:6e9:2b7:425, Dst: 2404:6800:4000:100c::64

Internet Control Message Protocol v6

HiPerConTracer Trace Service

0000 7a e3 b3 5c 3e 15

0010 51 9f 00 49 3a 46

0020 06 e9 02 b7 04 2f

0030 00 00 00 00 00 64

0040 1f 6a 00 00 00 00

0050 12 13 14 15 16 17

0060 22 23 24 25 26 27

0070 32 33 34 35 36 37

Internet Control Message Protocol v6: Protocol

Packets: 20685 - Displayed: 18 (0.1%) - Dropped: 0 (0.0%)

Profile: Default

## 5.4 TLS/HTTPS

- *Purpose:*

TLS provides encrypted communication over networks.

- *Example Observed:*

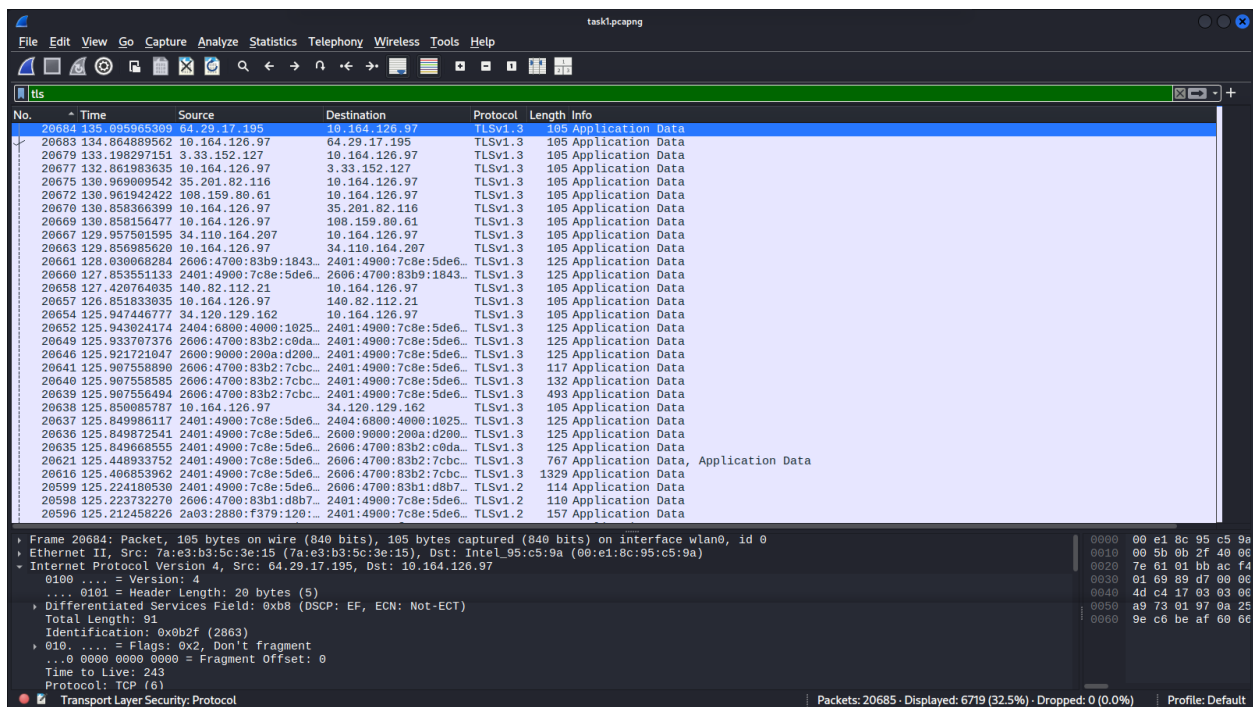
HTTPS traffic was generated while browsing websites.

- *Observation:*

Most web communication was encrypted and visible as TLS packets within Wireshark.

- *Packet Count:*

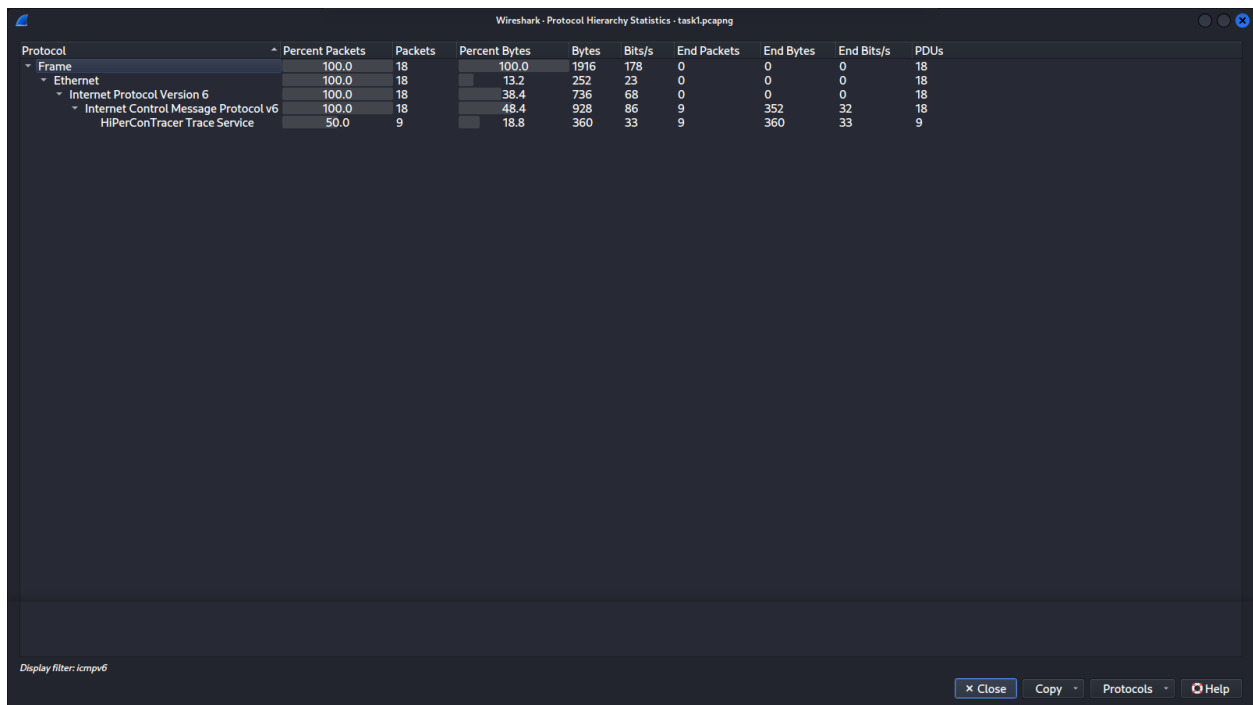
6719 Packets



## 6. Packet Statistics

The Protocol Hierarchy Statistics feature in Wireshark was used to analyze captured traffic.

| Protocol | Number of Packets |
|----------|-------------------|
| TCP      | 18353             |
| UDP      | 2304              |
| DNS      | 214               |
| ICMP     | 18                |
| TLS      | 6719              |





## 7. Packet Analysis

### DNS Analysis

The DNS protocol was responsible for resolving domain names before web connections were established.

Example:

Query: google.com

Response: Returned the corresponding IPv4 address.

Observation:

DNS communication occurred before HTTPS traffic began.

The image shows a Wireshark packet capture window titled 'task1.pcapng'. The 'dns' filter is applied, showing a list of DNS packets. The selected packet is packet 138, a standard query response from 10.164.126.97 to 10.164.126.36. The packet details pane shows the following structure:

- Frame 136: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlan0, id 0
- Ethernet II, Src: Intel\_95:c5:9a (00:e1:8c:95:c5:9a), Dst: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15)
- Internet Protocol Version 4, Src: 10.164.126.97, Dst: 10.164.126.36
- 0100 .... = Version: 4
- .... 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 56
- Identification: 0x9aad (39597)
- 010. .... = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header Checksum: 0x8e3a [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.164.126.97
- Destination Address: 10.164.126.36
- [Stream Index: 0]
- User Datagram Protocol, Src Port: 46899, Dst Port: 53
- Source Port: 46899
- Destination Port: 53
- Length: 36
- Checksum: 0xb6e6 [unverified]
- [Checksum Status: Unverified]
- [Stream Index: 1]
- [Stream Packet Number: 2]
- [Timestamps]
- UDP payload (28 bytes)
- Domain Name System (query)
- Transaction ID: 0x6bb5
- Flags: 0x0100 Standard query
- Questions: 1
- Answer RRs: 0
- Authority RRs: 0
- Additional RRs: 0
- Queries
- [Response In: 138]

The packet bytes pane shows the raw data in hexadecimal and ASCII. The status bar at the bottom indicates 'Packets: 20685 - Displayed: 214 (1.0%) - Dropped: 0 (0.0%)' and 'Profile: Default'.

## TCP Analysis

TCP was the dominant protocol observed during packet capture.

Observation:

- Three-way handshakes were visible.
- Reliable packet delivery mechanisms were observed.
- TCP was used by HTTPS applications.

The image shows a Wireshark packet capture window titled 'task1.pcapng'. The packet list pane at the top shows several packets, with packet 112 selected. The packet details pane shows the structure of the selected packet, and the packet bytes pane shows the raw data in hexadecimal and ASCII.

**Packet List:**

| No. | Time        | Source                | Destination           | Protocol | Length | Info                                                                                |
|-----|-------------|-----------------------|-----------------------|----------|--------|-------------------------------------------------------------------------------------|
| 111 | 6.282438381 | 2401:4900:7c8e:5de6:: | 2606:4700:83b2:7cbc:: | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=2045 Win=61 Len=0 TSval=1856620142 TSecr=4152926837 |
| 112 | 6.282439372 | 2401:4900:7c8e:5de6:: | 2606:4700:83b2:7cbc:: | TCP      | 86     | 37092 → 443 [ACK] Seq=21810 Ack=2111 Win=61 Len=0 TSval=1856620142 TSecr=4152926837 |
| 131 | 7.613440616 | 2401:4900:7c8e:5de6:: | 2606:4700:83b1:d8b7:: | TLSv1.2  | 174    | Application Data                                                                    |
| 132 | 7.814886557 | 2606:4700:83b1:d8b7:: | 2401:4900:7c8e:5de6:: | TCP      | 86     | 443 → 55996 [ACK] Seq=184 Ack=205 Win=18 Len=0 TSval=3448746836 TSecr=3482503816    |

**Packet Details (Packet 112):**

- Frame 112: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlan0, id 0
- Ethernet II, Src: Intel\_95:c5:9a (00:e1:8c:95:c5:9a), Dst: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15)
- Internet Protocol Version 6, Src: 2401:4900:7c8e:5de6:d171:6e9:2b7:425, Dst: 2606:4700:83b2:7cbc:c2fd:662:5ff2:75c4
- 0110 .... = Version: 6
- .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
- .... 0101 0000 1010 1011 1100 = Flow Label: 0x50abc
- Payload Length: 32
- Next Header: TCP (6)
- Hop Limit: 64
- Source Address: 2401:4900:7c8e:5de6:d171:6e9:2b7:425
- Destination Address: 2606:4700:83b2:7cbc:c2fd:662:5ff2:75c4
- [Stream index: 3]
- Transmission Control Protocol, Src Port: 37092, Dst Port: 443, Seq: 21810, Ack: 2111, Len: 0
- Source Port: 37092
- Destination Port: 443
- [Stream index: 2]
- [Stream Packet Number: 47]
- [Conversation completeness: Incomplete (12)]
- [TCP Segment Len: 0]
- Sequence Number: 21810 (relative sequence number)
- Sequence Number (raw): 61649410
- [Next Sequence Number: 21810 (relative sequence number)]
- Acknowledgment Number: 2111 (relative ack number)
- Acknowledgment number (raw): 1985282568
- 1000 .... = Header Length: 32 bytes (8)
- Flags: 0x010 (ACK)
- Window: 61
- [Calculated window size: 61]
- [Window size scaling factor: -1 (unknown)]
- Checksum: 0xb385 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- [Timestamps]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]

**Packet Bytes:**

0000 7a e3 b3 5c 3e 15  
0010 0a bc 00 29 06 4e  
0020 06 e9 02 b7 04 2e  
0030 06 02 5f f2 75 c4  
0040 02 08 00 10 00 3d  
0050 c6 0e f7 88 a2 75

Transmission Control Protocol: Protocol | Packets: 20685 - Displayed: 18353 (88.7%) - Dropped: 0 (0.0%) | Profile: Default

# ICMP Analysis

ICMP traffic was intentionally generated using the ping utility.

Example Command: ping google.com

Observation:

- Echo Request packets were sent.
- Echo Reply packets were received successfully.
- No packet loss was observed.

The image shows a Wireshark packet capture window titled 'task1.pcapng'. The packet list on the left shows several ICMPv6 Echo (ping) requests and replies. The selected packet is packet 139, an ICMPv6 Echo (ping) request. The packet details pane on the right shows the structure of the packet, including the Ethernet II header, Internet Protocol Version 6 header, and Internet Control Message Protocol v6 header. The packet bytes pane on the right shows the raw data of the packet.

Packet List:

| No.   | Time         | Source                 | Destination            | Protocol | Length | Info                                                                                           |
|-------|--------------|------------------------|------------------------|----------|--------|------------------------------------------------------------------------------------------------|
| 139   | 13.437518386 | 2401:4900:7c8e:5de6... | 2404:6800:4000:100c... | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=1, hop limit=64 (reply in 140) (SendTTL=20, Round=21)       |
| 140   | 13.216196395 | 2404:6800:4000:100c... | 2401:4900:7c8e:5de6... | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=1, hop limit=111 (request in 139) (SendTTL=20, Round=21)      |
| 143   | 14.138686462 | 2401:4900:7c8e:5de6... | 2404:6800:4000:100c... | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 144) (SendTTL=20, Round=21)       |
| 144   | 14.223116000 | 2404:6800:4000:100c... | 2401:4900:7c8e:5de6... | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=2, hop limit=111 (request in 143) (SendTTL=20, Round=21)      |
| 145   | 15.140397418 | 2401:4900:7c8e:5de6... | 2404:6800:4000:100c... | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=3, hop limit=64 (reply in 146) (SendTTL=20, Round=21)       |
| 146   | 15.237755306 | 2404:6800:4000:100c... | 2401:4900:7c8e:5de6... | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=3, hop limit=111 (request in 145) (SendTTL=20, Round=21)      |
| 147   | 16.142156948 | 2401:4900:7c8e:5de6... | 2404:6800:4000:100c... | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=4, hop limit=64 (reply in 148) (SendTTL=20, Round=21)       |
| 148   | 16.241715846 | 2404:6800:4000:100c... | 2401:4900:7c8e:5de6... | ICMPv6   | 118    | Echo (ping) reply id=0x0001, seq=4, hop limit=111 (request in 147) (SendTTL=20, Round=21)      |
| 149   | 17.144192895 | 2401:4900:7c8e:5de6... | 2404:6800:4000:100c... | ICMPv6   | 118    | Echo (ping) request id=0x0001, seq=5, hop limit=64 (no response found!) (SendTTL=20, Round=21) |
| 1648  | 54.382689970 | fe80::78e3:b3ff:fe5... | 2401:4900:7c8e:5de6... | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |
| 1649  | 54.382734791 | 2401:4900:7c8e:5de6... | fe80::78e3:b3ff:fe5... | ICMPv6   | 78     | Neighbor Advertisement 2401:4900:7c8e:5de6:d171:6e9:2b7:425 (sol)                              |
| 8620  | 78.476197418 | fe80::78e3:b3ff:fe5... | 2401:4900:7c8e:5de6... | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |
| 8630  | 78.476266126 | 2401:4900:7c8e:5de6... | fe80::78e3:b3ff:fe5... | ICMPv6   | 78     | Neighbor Advertisement 2401:4900:7c8e:5de6:d171:6e9:2b7:425 (sol)                              |
| 11535 | 85.341460730 | fe80::78e3:b3ff:fe5... | ff02::1                | ICMPv6   | 142    | Router Advertisement from 7a:e3:b3:5c:3e:15                                                    |
| 11536 | 85.351568445 | fe80::45a:b5ae:2fb5... | ff02::1                | ICMPv6   | 110    | Multicast Listener Report Message v2                                                           |
| 11572 | 85.754783297 | fe80::45a:b5ae:2fb5... | ff02::1                | ICMPv6   | 110    | Multicast Listener Report Message v2                                                           |
| 18852 | 98.931843473 | fe80::78e3:b3ff:fe5... | 2401:4900:7c8e:5de6... | ICMPv6   | 86     | Neighbor Solicitation for 2401:4900:7c8e:5de6:d171:6e9:2b7:425 from 7a:e3:b3:5c:3e:15          |

Packet Details:

Frame 140: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface wlan0, id 0

Ethernet II, Src: 7a:e3:b3:5c:3e:15 (7a:e3:b3:5c:3e:15), Dst: Intel\_95:c5:9a (00:e1:8c:95:c5:9a)

Internet Protocol Version 6, Src: 2404:6800:4000:100c::64, Dst: 2401:4900:7c8e:5de6:d171:6e9:2b7:425

0110 .... = Version: 6

.... 0011 0000 .... = Traffic Class: 0xb0 (DSCP: EF, ECN: Not-ECT)

.... 0011 0101 0001 1111 = Flow Label: 0x3519f

Payload Length: 64

Next Header: ICMPv6 (58)

Hop Limit: 111

Source Address: 2404:6800:4000:100c::64

Destination Address: 2401:4900:7c8e:5de6:d171:6e9:2b7:425

[Stream index: 4]

Internet Control Message Protocol v6

Type: Echo (ping) reply (129)

Code: 0

Checksum: 0xb0fd [correct]

[Checksum Status: Good]

Identifier: 0x0001

Sequence: 1

[Response To: 139]

[Response Time: 72.678 ms]

Timestamp from Echo data: Jun 3, 2026 13:14:30.819341000 IST

[Timestamp from Echo data (relative): 72.738105 milliseconds]

HiPerConTracer Trace Service

Internet Control Message Protocol v6: Protocol

Packets: 20685 - Displayed: 18 (0.1%) - Dropped: 0 (0.0%)

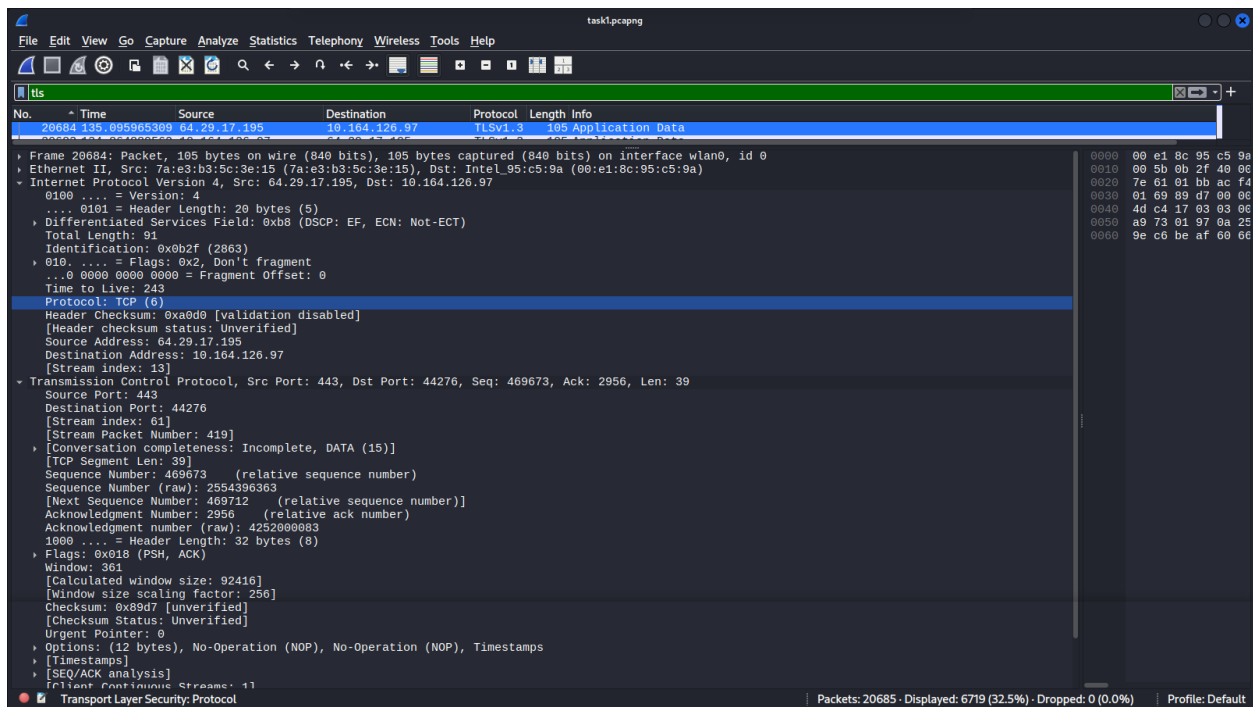
Profile: Default

## HTTPS/TLS Analysis

Encrypted communication was observed when browsing websites.

Observation:

- Packet contents could not be directly viewed because they were encrypted.
- TLS handshakes were visible.
- Destination domains included Google, GitHub, and YouTube.



## **8. Key Findings**

1. DNS queries occur before most web communications.
2. TCP was the most frequently observed protocol.
3. ICMP is useful for connectivity testing and diagnostics.
4. Modern websites primarily use encrypted HTTPS communication.
5. Wireshark can reveal extensive network information even when traffic is encrypted.

## **9. Challenges Faced**

- Large numbers of packets were generated during browsing activities.
- Identifying relevant packets required the use of protocol filters.
- Encrypted HTTPS traffic limited visibility into application-layer content.

## **10. Learning Outcomes**

Through this exercise, the following concepts were learned:

- Packet capture techniques.
- Network protocol analysis.
- DNS resolution process.
- TCP communication mechanisms.
- ICMP diagnostic functionality.
- Wireshark filtering and analysis techniques.
- Basic network troubleshooting methodology.

## **11. Conclusion**

This task successfully demonstrated the use of Wireshark for network traffic analysis. Multiple protocols including DNS, TCP, ICMP, and TLS were captured and analyzed. The exercise provided practical experience in packet capture, protocol identification, and network troubleshooting. Understanding packet-level communication is essential for cybersecurity professionals, network administrators, and penetration testers.

The analysis showed how different protocols interact to provide network connectivity and secure communication. Wireshark proved to be an effective tool for monitoring, analyzing, and understanding network traffic.

## **References**

1. Wireshark Official Documentation – <https://www.wireshark.org/docs>
2. Wireshark Wiki – <https://wiki.wireshark.org>
3. Cisco Networking Documentation – <https://www.cisco.com>